

Name of Student: _____

Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering (CSE)

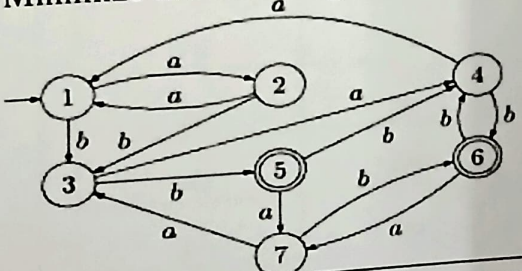
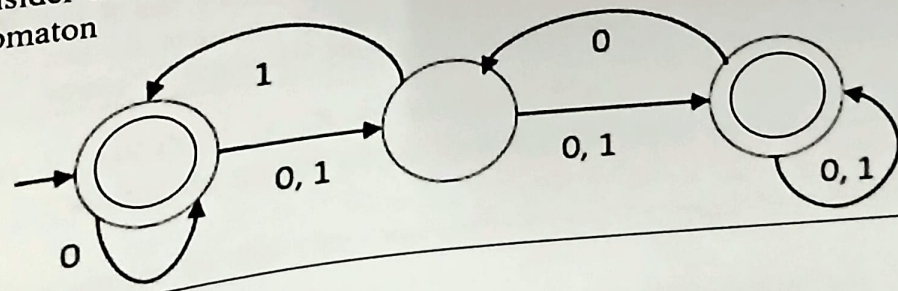
Mid-Term Examination March - 2025

Subject Name: Formal Language & Automata Theory
MM:20

Subject Code: CSE-225
Time Allotted: 90 min

NOTE: 1. All questions are compulsory.

2. Mark answer number clearly.

1.	Find DFA for the following regular languages: (a) All the strings of $\{0, 1\}$ begin with 01 and end with 01. (b) $L = \{(n_a(w) + 2n_b(w)) \bmod 3 > 2 \mid w \in \{a, b\}^*\}$	4
2.	Minimize the following DFA 	4
3.	Covert the following left linear regular grammar to equivalent right linear grammar $S \rightarrow Sab / Aa, A \rightarrow Bb / Aa / \epsilon, B \rightarrow Cba / d, C \rightarrow Ab / Ba / \epsilon$	4
4.	Construct a DFA equivalent to the following regular expressions $((aa + bb + \epsilon)(ab + ba)^* + a)^* + a$	4
5.	Consider the DFA M shown below find regular expression accepted by this automaton 	4